

```
!pip install transformers datasets

from transformers import GPT2LMHeadModel, GPT2Tokenizer
from transformers import DataCollatorForLanguageModeling
from transformers import Trainer, TrainingArguments
from datasets import load_dataset

# Create dataset
with open("my_dataset.txt", "w") as f:
    f.write("""
The sun rose over the mountains casting golden light across the valley.
The birds began to sing as the morning mist slowly faded away.
She walked through the forest listening to the rustling of leaves.
Every tree told a story of centuries gone by.
The river flowed gently carrying with it the memories of the land.
""")

tokenizer = GPT2Tokenizer.from_pretrained("gpt2")
model = GPT2LMHeadModel.from_pretrained("gpt2")
tokenizer.pad_token = tokenizer.eos_token

dataset = load_dataset("text", data_files={"train": "my_dataset.txt"})

def tokenize_function(examples):
    return tokenizer(examples["text"], truncation=True, max_length=64, padding="max_length")

tokenized_dataset = dataset.map(tokenize_function, batched=True, remove_columns=["text"])
data_collator = DataCollatorForLanguageModeling(tokenizer=tokenizer, mlm=False)

training_args = TrainingArguments(
    output_dir="./gpt2-finetuned",
    max_steps=100,
    per_device_train_batch_size=4,
    logging_steps=10,
    prediction_loss_only=True,
    learning_rate=5e-5,
)

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenized_dataset["train"],
    data_collator=data_collator,
)

trainer.train()
print("Training complete!")
```

Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
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 Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.23.0)
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 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->datasets) (1.17.0)
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Loading weights: 100%

148/148 [00:00<00:00, 613.80it/s, Materializing param=transformer.wte.weight]

GPT2LMHeadModel LOAD REPORT from: gpt2

Key	Status	
h.{0...11}.attn.bias	UNEXPECTED	

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

Generating train split: 300/0 [00:00<00:00, 18415.72 examples/s]

Map: 100%

300/300 [00:00<00:00, 6698.99 examples/s]

`loss\_type=None` was set in the config but it is unrecognized. Using the default loss: `ForCausalMLoss`.

[100/100 00:25, Epoch 1/2]

Step Training Loss

10	2.660720
20	0.611811
30	0.306032
40	0.282304
50	0.283446
60	0.240389
70	0.136656
80	0.170809
90	0.227598
100	0.200532

Writing model shards: 100%

1/1 [00:07<00:00, 7.81s/it]

Training complete!

```

from transformers import pipeline

generator = pipeline("text-generation", model=model, tokenizer=tokenizer)

prompt = "The sun rose over"

result = generator(
    prompt,
    max_new_tokens=50,
    num_return_sequences=3,
    temperature=0.9,
    do_sample=True,
)

for i, r in enumerate(result):
    print(f"\n--- Output {i+1} ---")
    print(r["generated_text"])

```

Passing `generation\_config` together with generation-related arguments=(`max\_new\_tokens`, `num\_return\_sequences`, `temperature`) Both `max\_new\_tokens` (=50) and `max\_length` (=50) seem to have been set. `max\_new\_tokens` will take precedence. Please refer

```

--- Output 1 ---
The sun rose over the mountains casting golden light across the valley. The morning mist slowly faded away.

Soon the morning mist slowly faded away. The memories of the land slowly faded away. The memories of the land slowly faded a

The land was filled

--- Output 2 ---
The sun rose over the mountains casting golden light across the valley.

The morning mist gradually faded away as the hours passed by.

It was the morning mist that had been the memories of the land.

The memories of the land were slowly taking root

--- Output 3 ---
The sun rose over the mountains casting golden light across the valley. A tear began to fall down the valley as the memories

It took a moment to get used to the memories of the land had brought.

The next morning

```

```

from transformers import GPT2LMHeadModel, GPT2Tokenizer, pipeline

```

```

# Load original GPT-2
tokenizer = GPT2Tokenizer.from_pretrained("gpt2")
model = GPT2LMHeadModel.from_pretrained("gpt2")
tokenizer.pad_token = tokenizer.eos_token

print("Model loaded!")

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.

```

warnings.warn(
tokenizer_config.json: 100%                26.0/26.0 [00:00<00:00, 528B/s]

vocab.json:      1.04M/? [00:00<00:00, 1.61MB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
merges.txt:      456k/? [00:00<00:00, 927kB/s]

tokenizer.json:  1.36M/? [00:00<00:00, 4.47MB/s]

config.json: 100%                665/665 [00:00<00:00, 65.0kB/s]

model.safetensors: 100%          548M/548M [00:03<00:00, 335MB/s]

Loading weights: 100%            148/148 [00:00<00:00, 525.27it/s, Materializing param=transformer.wte.weight]

GPT2LMHeadModel LOAD REPORT from: gpt2
Key                | Status      | 
-----+-----+
h.{0...11}.attn.bias | UNEXPECTED | 

Notes:
- UNEXPECTED      :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
generation_config.json: 100%      124/124 [00:00<00:00, 7.49kB/s]

Model loaded!

```